Kianté Brantley
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Professional Experience

Harvard University

2024 – present Assistant Professor Kempner Institute and School of Engineering and Applied Sciences

Cornell University

2022 – 2024 Postdoctoral Associate, Department of Computer Science

Microsoft Research

2021 Research Intern, Redmond
2020 Research Intern, Montréal
2019 Research Intern, NYC
2018 Research Intern, NYC

US Department of Defense

2015 – 2017 Cyber Security Data Scientist
2011 – 2015 Cyber Security Software Engineer
2010 Software Engineer Intern

Education

University of Maryland, College Park

2016 – 2022 Ph.D., Computer Science
Advisor: Hal Daumé III.
Thesis title: Expert-In-The-Loop for Sequential Decisions and Predictions

New York University, Center for Data Science

2018 – 2022 Visiting Researcher, Computer Science Host: Kyunghyun Cho

University of Maryland, Baltimore County

2015 – 2016 M.Sc. Computer Science
Thesis title: BCAP: A Pruning Technique to Reduce Over-fitting

2013 – 2015 B.Sc. Computer Science
Minor in Mathematics

The Community College of Baltimore County

2011 – 2013 A.A. Computer Science and Mathematics

Honors and Awards

2023 Spotlight Talk ICLR
2022 NSF Computing Innovation Fellowship (CIFellows)
2021 Spotlight talk, Empirical Methods in Natural Language Processing
2021 Spotlight talk, International Conference on Learning Representations

2021 Spotlight talk, International Conference on Learning Representations
2021 2nd place, The New York Academy of Sciences Star Talks
2021 Ann G. Wylie Dissertation Fellowship
2020 Microsoft Dissertation Research Grant
2019 Spotlight Talk ICLR
2018 ACM SIGHPC/Intel Computational and Data Science Fellowship
2017 Sloan Research Fellowship
2016 Louis Stokes Alliance for Minority Participation Bridge to the Doctorate Program Fellowship
2016 University of Maryland College Park Dean's Fellowship
2015 Department of Defense Graduate Fellowship
2013 Transfer-Scholarships in Information Technology and Engineering (T-SITE) Scholar
2013 Maryland Senators and Delegates Scholarship
2013 Transfer Student Alliance Scholarship
2013 Howard P. Rawlings Educational Assistance Grant
2013 CCBC Foundation General Scholarship

Publications

Thesis

1. **Brantley, K.** "Expert-in-the-Loop for Sequential Decisions and Predictions". PhD thesis. University of Maryland, College Park, 2021.
2. **Brantley, K.** "BCAP: An Artificial Neural Network Pruning Technique to Reduce Overfitting". MA thesis. University of Maryland, Baltimore County, 2016.

Conference Papers

1. **Brantley, K.**, Fang, Z., Dean, S., Joachims, T., "Ranking with Long-Term Constraints". In: *Proceedings of the ACM International Conference on Web Search and Data Mining (WSDM)*. 2024, pp. 47–56.
2. Chang, J. D., Sreenivas, D., Huang, Y., **Brantley, K.**, Sun, W., "Adversarial Imitation Learning via Boosting". In: *The International Conference on Learning Representations (ICLR)*. 2024.
3. Gao, Z., Chang, J. D., Zhan, W., Oertell, O., Swamy, G., **Brantley, K.**, Joachims, T., Bagnell, J. A., Lee, J. D., Sun, W., "REBEL: Reinforcement Learning via Regressing Relative Rewards". In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2024, pp. 52354–52400.
4. Gao, Z., Zhan, W., Chang, J. D., Swamy, G., **Brantley, K.**, Lee, J. D., Sun, W., "Regressing the Relative Future: Efficient Policy Optimization for Multi-turn RLHF". In: *Proceedings of the International Conference on Machine Learning (ICML)*. 2024.
5. Monea, G., Bosselut, A., **Brantley, K.**, Artzi, Y., "LLMs Are In-Context Reinforcement Learners". In: *Conference On Language Modeling*. 2024.
6. Oertell, O., Chang, J. D., Zhang, Y., **Brantley, K.**, Sun, W., "RL for Consistency Models: Faster Reward Guided Text-to-Image Generation". In: *Reinforcement Learning Conference*. 2024.
7. Phan, M., **Brantley, K.**, Milani, S., Mehri, S., Swamy, G., Gordon, G. J., "When is Transfer Learning Possible?" In: *Proceedings of the International Conference on Machine Learning (ICML)*. 2024.
8. Tucker, A. D., **Brantley, K.**, Cahall, A., Joachims, T., "Coactive Learning for Large Language Models using Implicit User Feedback". In: *Proceedings of the International Conference on Machine Learning (ICML)*. 2024.
9. Wu, R., Chen, Y., Swamy, G., **Brantley, K.**, Sun, W., "Diffusing States and Matching Scores: A New Framework for Imitation Learning". In: *The International Conference on Learning Representations (ICLR)*. 2024.
10. Faltings, F., Galley, M., **Brantley, K.**, Peng, B., Cai, W., Zhang, Y., Gao, J., Dolan, W. B., "Interactive Text Generation". In: *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2023, pp. 4450–4468.

11. Ramamurthy, R., Ammanabrolu, P., **Brantley, K.**, Hessel, J., Sifa, R., Bauckhage, C., Hajishirzi, H., Choi, Y., "Is Reinforcement Learning (Not) for Natural Language Processing: Benchmarks, Baselines, and Building Blocks for Natural Language Policy Optimization". In: *The International Conference on Learning Representations (ICLR)*. Spotlight. 2023.
12. Wu, A., **Brantley, K.**, Kojima, N., Artzi, Y., "lilGym: Natural Language Visual Reasoning with Reinforcement Learning". In: *Proceedings of the Conference of the Association for Computational Linguistics (ACL)*. 2023.
13. **Brantley, K.**, Mehri, S., Gordon, G. J., "Successor feature sets: Generalizing successor representations across policies". In: *Proceedings of the National Conference on Artificial Intelligence (AAAI)*. Vol. 35. 13. 2021, pp. 11774–11781.
14. **Brantley, K.**, Dudik, M., Lykouris, T., Miryoosefi, S., Simchowitz, M., Slivkins, A., Sun, W., "Constrained episodic reinforcement learning in concave-convex and knapsack settings". In: *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. 33. 2020, pp. 16315–16326.
15. **Brantley, K.**, Sharaf, A., Daumé III, H., "Active Imitation Learning with Noisy Guidance". In: *Proceedings of the Conference of the Association for Computational Linguistics (ACL)*. 2020, pp. 2093–2105.
16. **Brantley, K.**, Sun, W., Henaff, M., "Disagreement-regularized imitation learning". In: *The International Conference on Learning Representations (ICLR)*. Spotlight. 2019.
17. Miryoosefi, S., **Brantley, K.**, Daumé III, H., Dudik, M., Schapire, R. E., "Reinforcement learning with convex constraints". In: *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. 32. 2019.
18. Welleck, S., **Brantley, K.**, Iii, H. D., Cho, K., "Non-monotonic sequential text generation". In: *Proceedings of the International Conference on Machine Learning (ICML)*. PMLR. 2019, pp. 6716–6726.

Workshop Papers

1. Chang, J., **Brantley, K.**, Ramamurthy, R., Misra, D., Sun, W., "Learning to Generate Better Than Your LLM". In: *NeurIPS 2023 Workshop on Instruction Tuning and Instruction Following*. 2023.
2. Gao, G., Chang, J. D., Cardie, C., **Brantley, K.**, Joachims, T., "Policy-Gradient Training of Language Models for Ranking". In: *NeurIPS 2023 Workshop Foundation Models for Decision Making Workshop*. 2023.
3. Sharaf, A., Feng, S., Nguyen, K., **Brantley, K.**, Daumé III, H., "The UMD Neural Machine Translation Systems at WMT17 Bandit Learning Task". In: *Proceedings of the Second Conference on Machine Translation*. 2017, pp. 667–673.
4. Ganesan, A., **Brantley, K.**, Pan, S., Chen, J., "Ldaexplore: Visualizing topic models generated using latent dirichlet allocation". In: *extvis Workshop - Intelligent User Interfaces (IUI)*. 2015.

Preprint Papers

1. **Brantley, K.**, Chen, M., Gao, Z., Lee, J. D., Sun, W., Zhan, W., Zhang, X., "Accelerating RL for LLM Reasoning with Optimal Advantage Regression". In: *Preprint*. 2025.
2. Espinosa-Dice, N., Zhang, Y., Chen, Y., Guo, B., Oertell, O., Swamy, G., **Brantley, K.**, Sun, W., "Scaling Offline RL via Efficient and Expressive Shortcut Models". In: *Preprint*. 2025.
3. Wang, K., Zhou, J. P., Chang, J., Gao, Z., Kallus, N., **Brantley, K.**, Sun, W., "Value-Guided Search for Efficient Chain-of-Thought Reasoning". In: *Preprint*. 2025.
4. Zhou, J. P., Wang, K., Chang, J., Gao, Z., Kallus, N., Weinberger, K. Q., **Brantley, K.**, Sun, W., "Q: Provably Optimal Distributional RL for LLM Post-Training". In: *Preprint*. 2025.
5. Chang, J. D., Shan, W., Oertell, O., **Brantley, K.**, Misra, D., Lee, J. D., Sun, W., "Dataset Reset Policy Optimization for RLHF". In: *Preprint*. 2024.
6. Gao, Z., **Brantley, K.**, Joachims, T., "Reviewer2: Optimizing Review Generation Through Prompt Generation". In: *Preprint*. 2024.
7. Wu, A., **Brantley, K.**, Artzi, Y., "A Surprising Failure? Multimodal LLMs and the NLVR Challenge". In: *Preprint*. 2024.

Funding

2022 – 2024 NSF Computing Innovation Fellowship (CIFellows), $255k
2021 Ann G. Wylie Dissertation Fellowship, $15k
2020 Microsoft Dissertation Research Grant, $25k
2018 – 2020 ACM SIGHPC/Intel Computational and Data Science Fellowships, $30k
2017 – 2020 Sloan Research Fellowship, $15k
2016 – 2018 Louis Stokes Alliance for Minority Participation Bridge to the Doctorate Program Fellowship, $80k
2016 – 2018 University of Maryland College Park Dean's Fellowship, $5k
2015 Department of Defense Graduate Fellowship, $40k

Invited Talks

The Power of Resets!

2025 CCC Computing Futures Symposium.
2025 Google.
2025 Berkeley Simons Institute: The Future of Language Models and Transformers.

Efficient Policy Optimization Techniques for LLMs

2025 NSF RL Workshop Panel.
2024 MIT.

Learning From Interaction

2024 George Washington University.
2024 Emory University.
2024 Stony Brook University.
2024 Columbia University.
2024 Rice University.
2024 Harvard University.
2024 Rutgers University.
2024 Princeton University.
2024 Microsoft Research - NYC.
2024 University of Michigan.
2024 Northeastern.
2024 University of Illinous Chicag.

Ranking with Long Term Constraints

2023 Wayfair.

Reinforcement Learning from Guided Feedback: Addressing the Shortcoming of RL in NLP

2023 Rochester Institute of Technology.
2023 NC State University.

Expert-in-the-Loop for Sequential Decisions and Prediction

2023 Cornell NLP Seminar.
2023 Google Brain Montreal.
2021 University of Maryland UMIACS.

Successor feature sets: Generalizing successor representations across policies

2021 Microsoft Research Summit.

Learning through interaction with experts

2020 Adobe Research.

Active Imitation Learning with Noisy Guidance

2020 The New York Academy of Sciences.

Non-monotonic Sequential Text Generation

2020 Black in AI Workshop at ACL.

Academic Service

Area Chair

2025 Conference of Language Modeling
2023 Empirical Methods in Natural Language Processing

Program Chair

Special Interest Group on Information Retrieval (SIGIR)
ICML Workshop Spurious correlations, Invariance, and Stability
International Conference on Learning Representations (ICLR)
European Workshop on Reinforcement Learning
Empirical Methods in Natural Language Processing (EMNLP)
Conference on Neural Information Processing Systems (Neurips)
European Chapter of the Association for Computational Linguistics (EACL)
Conference of the Association for Computational Linguistics (ACL)
International Conference on Machine Learning (ICML)
Tapia Scholarship Reviewer
Black in Ai Admission

Workshops

2021 – 2022 Co-chair, Interactive Learning for NLP workshop @ ACL and Neurips
2021 – 2022 Co-chair, NYU AI Winter School for underrepresented groups
2021 Co-chair, Black in Ai at AAAI

Co-curricular

2023 Participated, Cornell NextGen Professor
2022 Volunteered, Ithaca College Central NY LSAMP symposium
2022 Volunteered, WiNLP Panel at NAACL
2022 Volunteered, UMD Doctoral Career Pathways
2020 Participated, CMD-IT Academic Careers Workshop
2020 Volunteered, Bitview (teaching high schoolers computer science)
2020 Volunteered, Industry Mentor UMBC CWIT
2017 Volunteered, Maryland Institute for Minority Achievement and Urban Education College Conference
2015 – 2017 Volunteered, MSDE CTE/PLTW Conference Student Panel
2013 – 2016 Volunteered, Western Tech High Open house

Department and University Service

2025 Harvard Kempner Fellows Committee
2025 Harvard Faculty Committee

Teaching

Course Instructor

2025 Planning and Learning Methods in AI
2020 Bitview (teaching high students computer science)

Teaching Assistant

2019 Computational Linguistics 1

Miscellaneous Experience

Certifications

2013 **Object-Oriented Programming Certificate Community College of Baltimore County** .

Technical Reports

2017 Deep Reinforcement Learning for Sentence Compression
2017 Recurrent Neural Network for Sequence-to-Sequence Reinforcement Learning
2017 Experimental Comparison of Density-Based Spatial Clustering of Applications with Noise using k-d Trees vs. using VP Trees
2015 Comparing the effect of using PCA and DAE on classifier performance in predicting Epileptic Seizures from EEG data

Non-Research Employment

2010 Social Security Administration, Summer Intern
2010 Baltimore Convention Center, Waiter
2010 McDonald, Cashier
2007 – 2009 Green House Cage, Prep Cook